

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	S.Blessika
Register Number	192525251

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I S.Blessika(192525251(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Blessika

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim

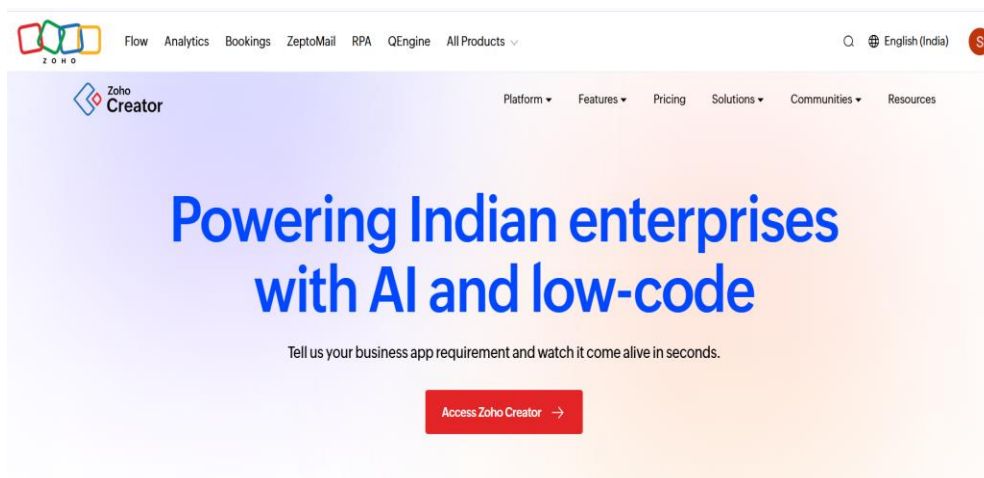
To create a simple cloud-based Library Book Reservation System using a Cloud Service Provider to demonstrate Software as a Service (SaaS) with details such as book title, author, publication, year, edition, number of copies, rack number, and book status.

Algorithm

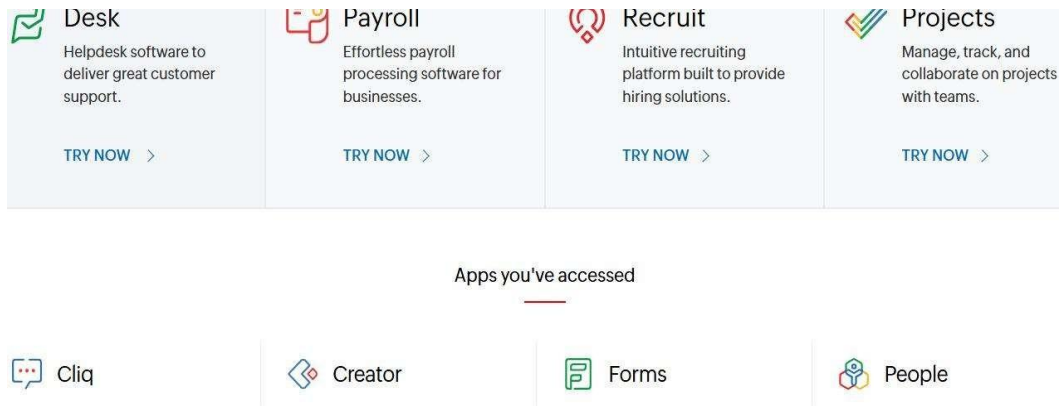
1. Open the selected cloud service provider.
2. Create a new cloud application.
3. Create a form named Library Book Reservation.
4. Add fields for book details.
5. Add fields for reservation and availability status.
6. Configure the form and save it.
7. Add sample book records.
8. Create a report/list to display the books.
9. Test the application by adding and updating book records.
10. Verify that the application works through the cloud.

Implementation

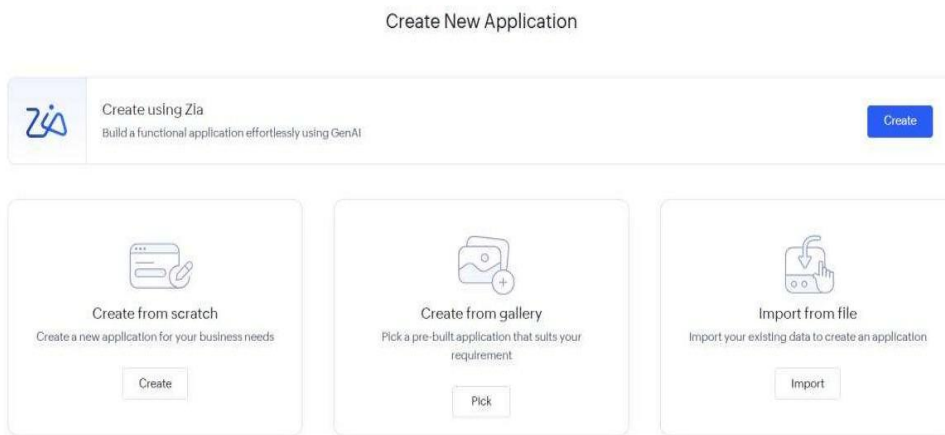
Step-1 : Open Zoho Creator



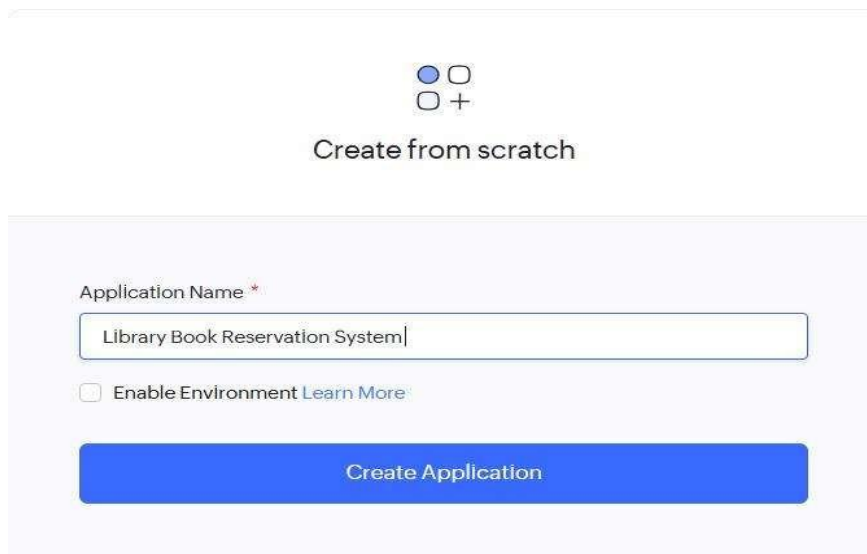
Step-2: Login to your Zoho account and open Zoho Creator



Step-3: Create new Application and Click on create From Scratch



Step-4: Enter Application Name As Library book Reservation system and click on Create applications



Step 5: Add fields : Book Title , Author ,Publication, Edition, Rack Number

Dashboard

Reservations

Books

Copy Inventory

Reservations

Book

User

Reservation Date

dd-MMM-yyyy

Due Date

dd-MMM-yyyy

Status

-Select-

Return Date

dd-MMM-yyyy

Created Time

dd-MMM-yyyy HH:mm:ss

Last Modified Time

dd-MMM-yyyy HH:mm:ss

Submit

Reset

BOOKS

Title

Author

Publication

Year

#####

Edition

Total Copies

#####

Rack Number

Status

-Select-

Created Time

dd-MMM-yyyy HH:mm:ss

Last Modified Time

dd-MMM-yyyy HH:mm:ss

Submit

Reset

Step-6: Check the uploaded fields in the reports

DashboardReservation RecordsBook Inventories

All Reservation Records

Q+...

Book	Member Name	Member Email	Member Phone	Reservation Date	Due Date	Return Date	Reservation Status	Issued
The Silent Echo Quantum Horizons Urban Gardening Guide Digital Art Revolution	Eva Brown	eva.b@example.com	+17533568986	08-Aug-2026 14:20:00	12-Aug-2026	12-Aug-2026	Cancelled	Sarah L
	David Kim	d.kim@example.com	+19897919456	08-Aug-2026 13:00:00	25-Aug-2026	25-Aug-2026	Active	Mike Ch
	Carol White	carol.w@example.com	+19074662914	08-Aug-2026 11:45:00	10-Aug-2026	10-Aug-2026	Active	Sarah L
	Bob Martinez	bob.m@example.com	+17268505021	08-Aug-2026 10:30:00	20-Aug-2026	20-Aug-2026	Returned	Mike Ch
The Silent Echo Quantum Horizons Midnight Jazz Club	Alice Johnson	alice.j@example.com	+19706118732	08-Aug-2026 09:15:00	15-Aug-2026	15-Aug-2026	Active	Sarah L

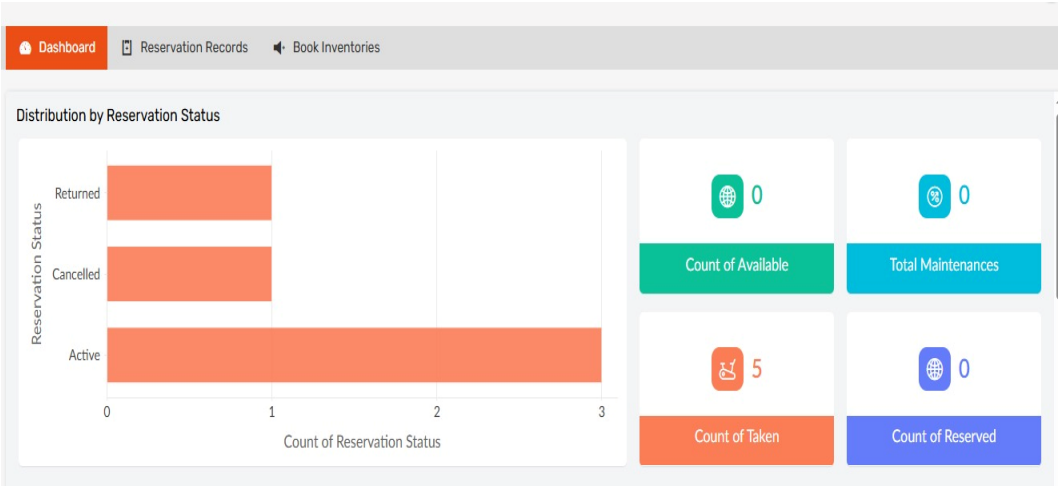
DashboardReservation RecordsBook Inventories

All Book Inventories

Q+...

Book Title	Author Name	Publisher Name	Publicatio...	Edition Number	Total C...	Rack Location	Book Status	ISBN Code
seethey	bosh	bosh	2022	3	50	3 row -2col		IS2859
Digital Art Revolution	Maya Lin	Tech Art Press	2026	1st	25	E-03	Taken	978-1-4843-5122-8
Midnight Jazz Club	Leo Rivera	Rhythm Records	2022	3rd	40	D-14	Taken	978-0-307-73662-4
Urban Gardening Guide	Sarah Green	Nature Books	2025	1st	30	C-08	Taken	978-1-5067-6134-2
Quantum Horizons	Dr. Aris Thorne	Science Today	2023	2nd	75	B-05	Taken	978-0-590-86362-5
The Silent Echo	Elena Vance	Global Press	2024	1st	50	A-12	Taken	978-3-16-148410-0

Step-7 : Application is created succesfully



Result

Thus, a **cloud-based Library Book Reservation System** was successfully created using SaaS, allowing library book details and reservation status to be managed through the cloud

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Aim

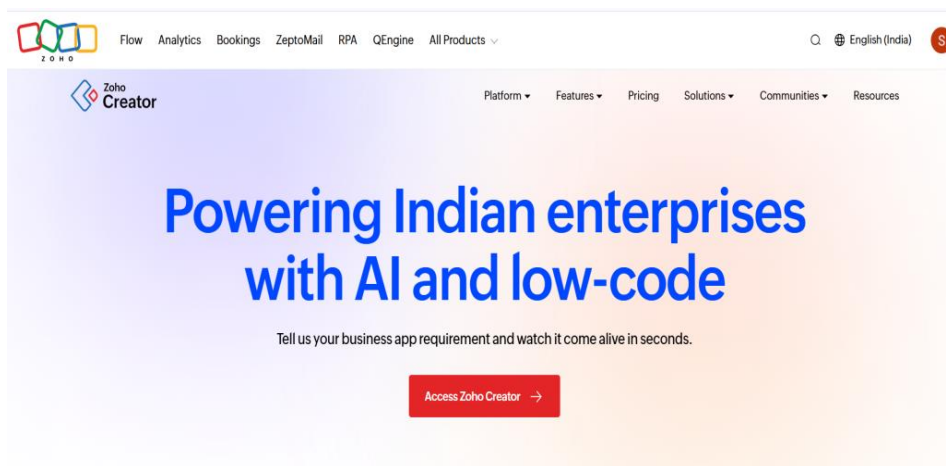
To create a simple **cloud-based Payroll Processing System** using a Cloud Service Provider to demonstrate **Software as a Service (SaaS)** and calculate employee salary based on basic pay, HRA, DA, TA, gross pay, and net pay.

Algorithm

1. Open the selected cloud service provider.
2. Create a new cloud application.
3. Create a form named **Employee Payroll**.
4. Add employee information fields.
5. Add salary-related fields.
6. Calculate allowances such as HRA, DA, and TA.
7. Calculate gross pay.
8. Calculate net pay.
9. Create a payroll report.
10. Enter sample employee records and verify the salary calculation.

Implementation

Step-1 : Open Zoho Creator



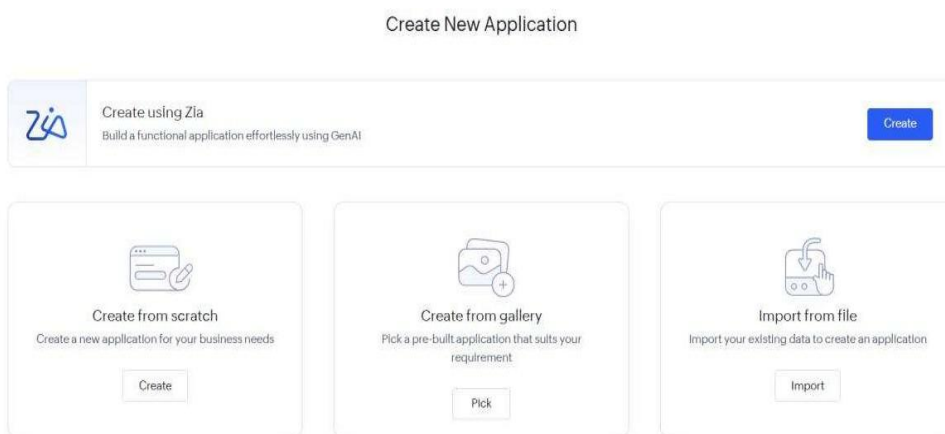
Step-2: Login to your Zoho account and open Zoho Creator



Apps you've accessed



Step-3: Create new Application and Click on create From Scratch



Step-4: Enter Application Name As Payroll processing system and click on Create applications

A form titled 'Create from scratch'. It features a text input field for 'Application Name' containing 'Payroll Processing System'. Below the input is a checkbox labeled 'Enable Environment' with a 'Learn More' link. At the bottom is a large blue button labeled 'Create Application'.

Step 5: Add fields

Payroll Processing System

Dashboard

Salary Componen...

+ Salary Component

All Salary Compon...

Periods

Employee Masters

Records

Salary Component

Component Name

Component Code

Amount

#####.##

☐ Is Taxable

Employee

-Select-

Pay Period

-Select-

Calculation Type

-Select-

Remarks

Payroll Processing System

Dashboard

Salary Componen...

Employee Masters

+ Employee Master

All Employee Mast...

Designations

Records

Employee Master

Employee Name

First Name

Total Experience

#####

Email Address

Phone Number

+91

81234 56789

Address

Address Line 1

Designation

-Select-

Department

-Select-

Employment Status

-Select-

Step-6: Check the uploaded fields in the reports

Payroll Processing System

Dashboard

Salary Componen...

+ Salary Component

All Salary Compon...

Periods

Employee Masters

Records

All Salary Components

Component Name

Component Code

Amount

Is Taxable

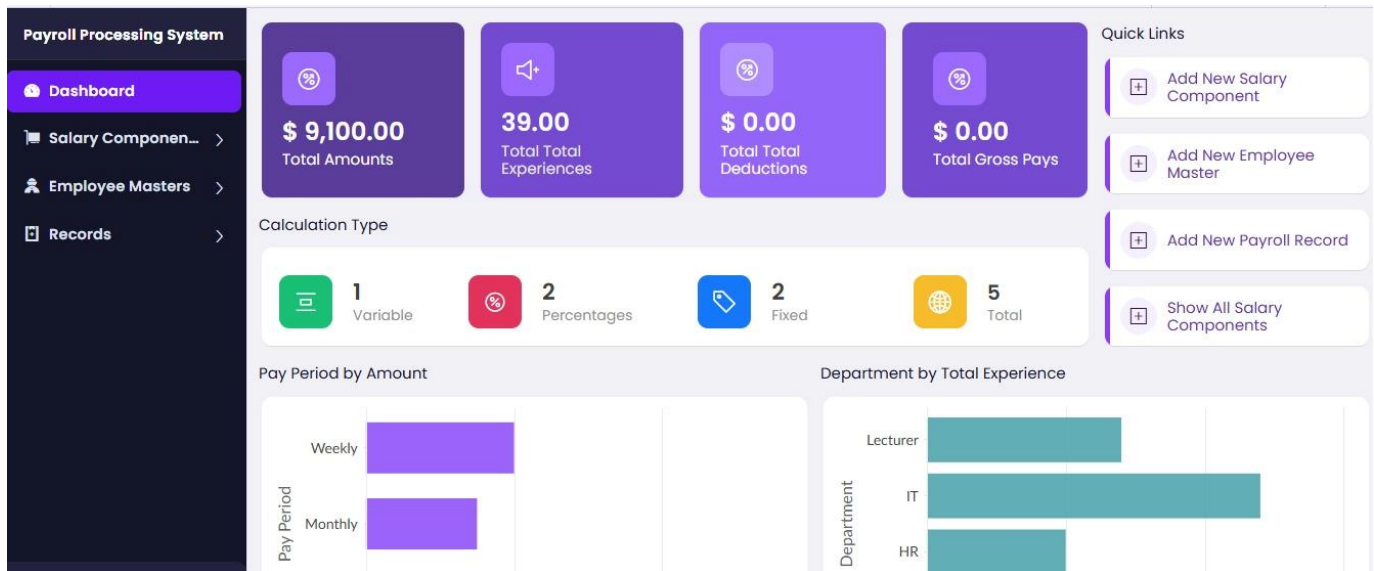
Employee

Pay Period

Calculation Type

Performance Bonus	PB005	\$ 1,200.00	false	Alice Johnson Robert Williams Michael Chen	Weekly	Fixed
Transport Allowance	TRP004	\$ 800.00	true	Alice Johnson Robert Williams	Daily	Fixed
Medical Insurance	MED003	\$ 800.00	false	Alice Johnson Robert Williams Sarah Martinez Michael Chen Emily Davis	Weekly	Percentage
House Rent Allowance	HRA002	\$ 1,500.00	true	Alice Johnson Robert Williams Sarah Martinez Michael Chen	Monthly	Variable
Basic Salary	BS001	\$ 5,000.00	true	Alice Johnson Robert Williams Sarah Martinez	Daily	Percentage

Step-7 : Application is created successfully



Result

Thus, a **cloud-based Payroll Processing System** was successfully created using SaaS to store employee details and automatically calculate **gross pay and net pay**.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim

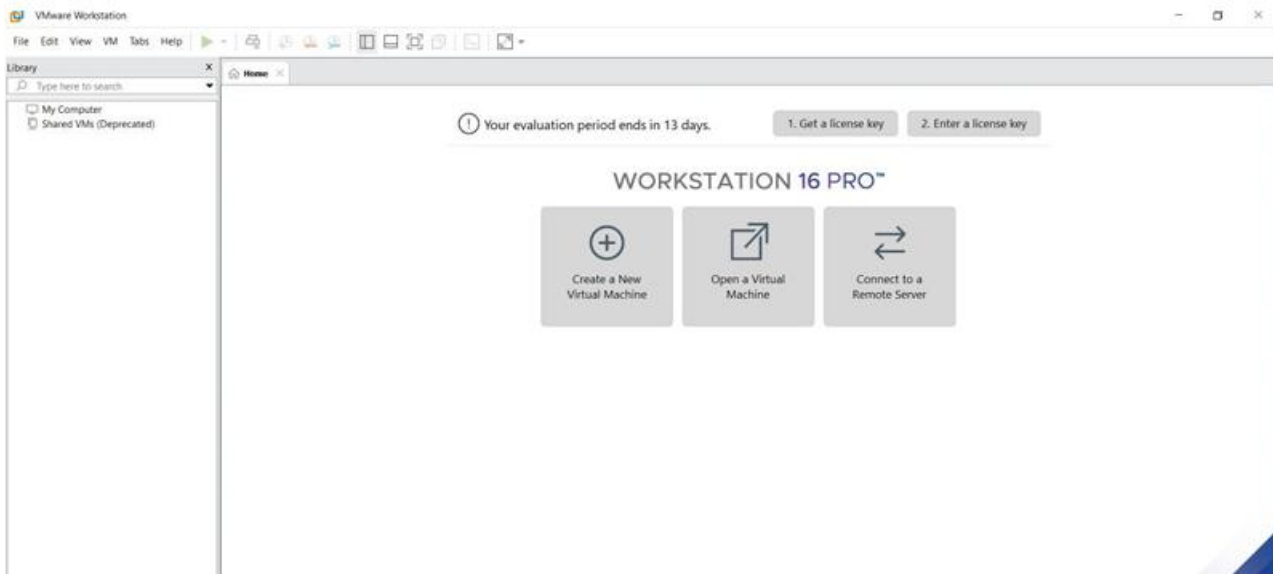
To demonstrate virtualization by installing a Type-2 Hypervisor (Oracle VirtualBox) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage using a Windows/Linux host operating system.

Algorithm

1. Install Oracle VirtualBox on the host operating system.
2. Download a Linux/Windows ISO image.
3. Open VirtualBox and create a new Virtual Machine.
4. Select the VM name and operating system type.
5. Allocate 2 GB RAM to the VM.
6. Allocate 1 CPU to the VM.
7. Create a virtual hard disk with 15 GB storage.
8. Attach the downloaded ISO image to the VM.
9. Start the Virtual Machine.
10. Install the guest operating system.
11. Verify that the VM starts and works correctly.

Implementation Steps

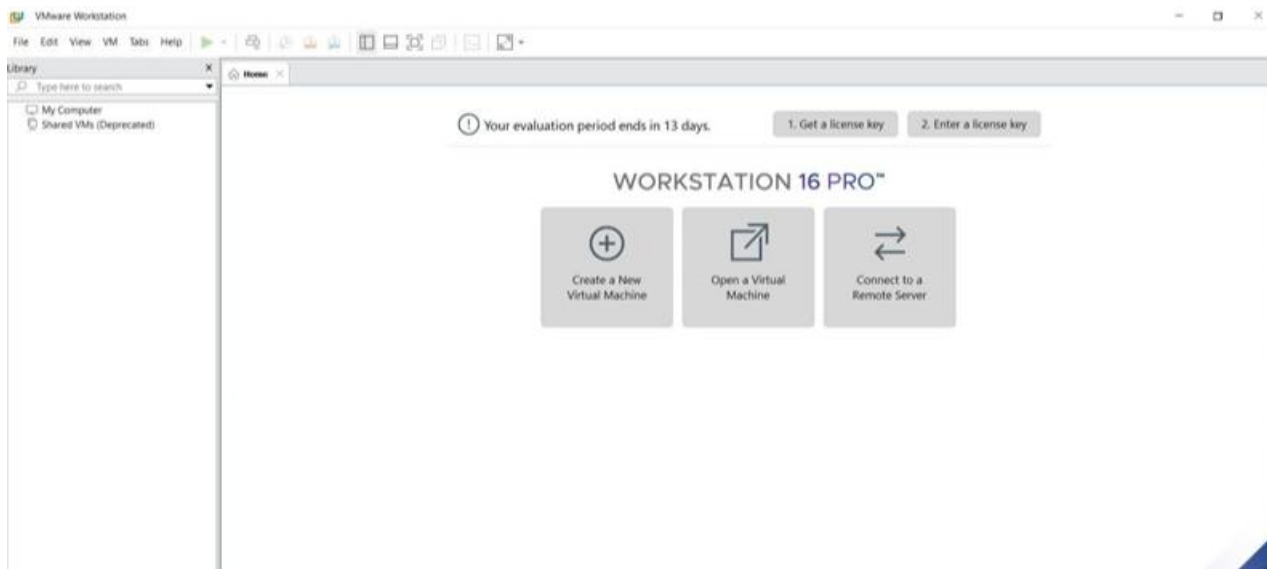
STEP 1: DOWNLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2 HYPERVISOR



Index of /11.x/x86/release/

distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.

 Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☐ Installer disc image file (iso):

C:\Users\Gopi Prasad-PC\Downloads\ubuntu-20.04.2

Browse...

☒ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help

< Back

Next >

Cancel

Connect to a Remote Server

New Virtual Machine Wizard

Select a Guest Operating System

Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESXi

☐ Other

Version

Ubuntu 64-bit

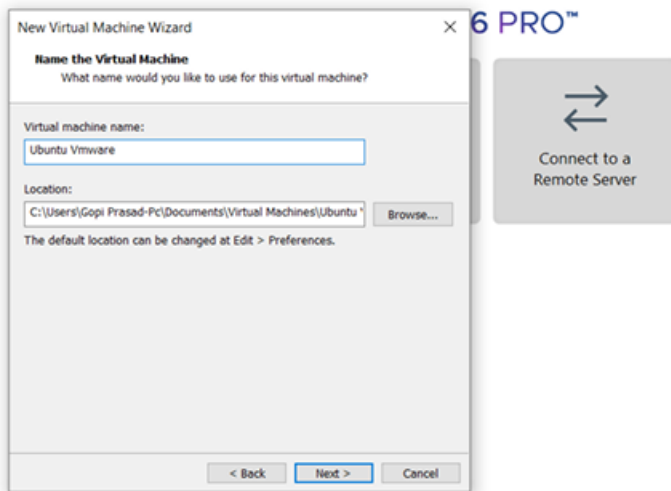
Help

< Back

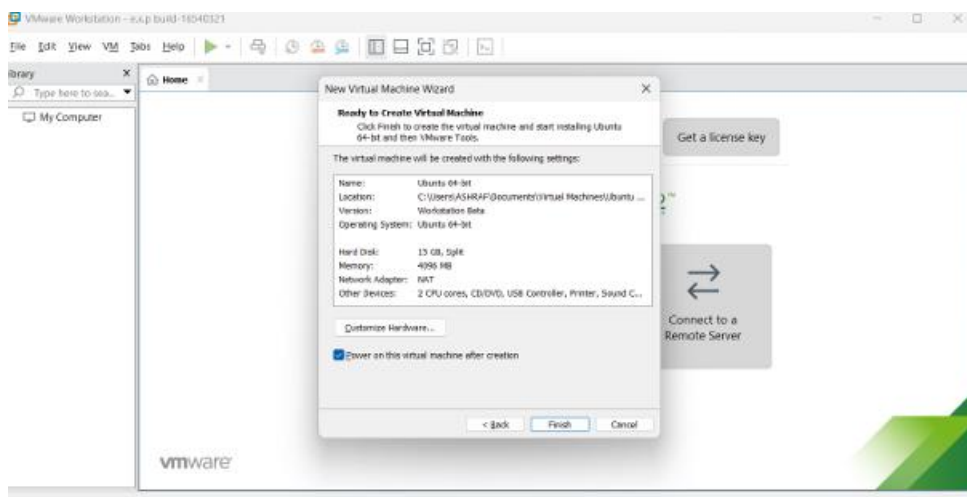
Next >

Cancel

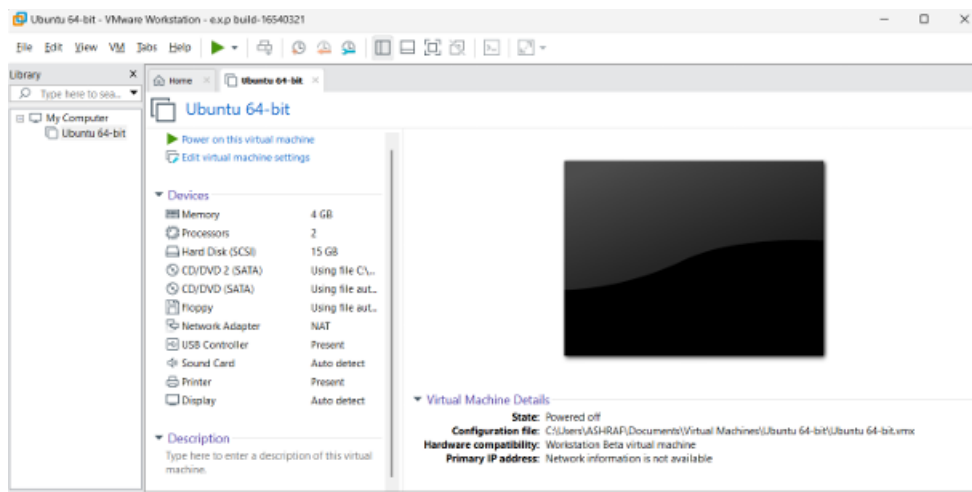
Connect to a Remote Server



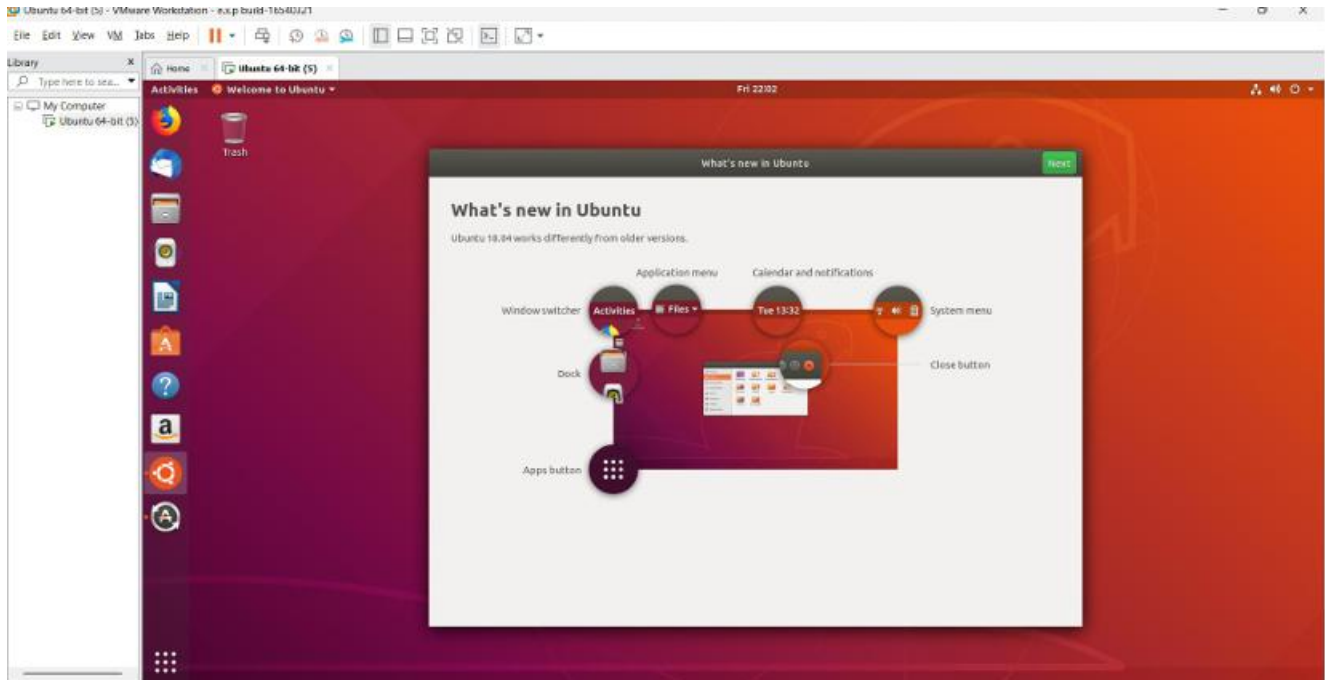
STEP 5: Set RAM = 2 GB and CPU = 1.



STEP 6: Set the hard disk size = 15 GB.



STEP 7: Start the VM and install Ubuntu



Result : The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

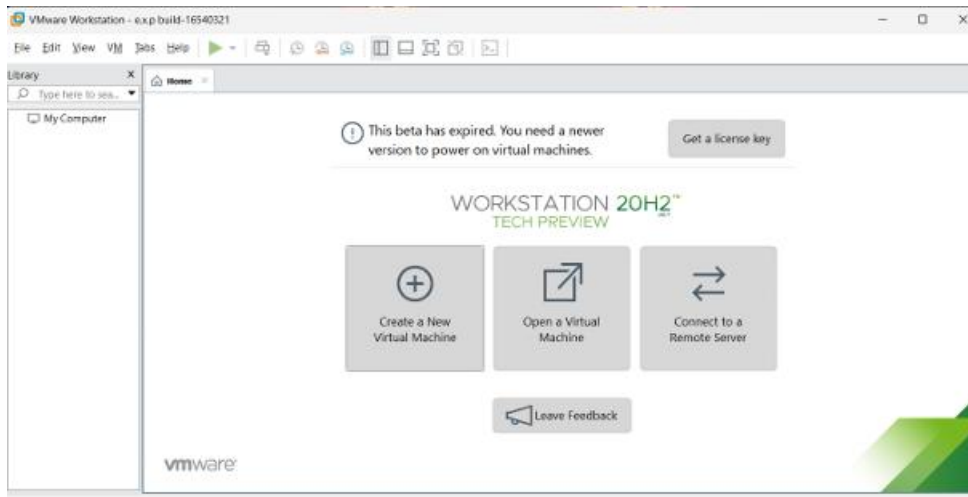
To demonstrate virtualization by creating a **clone of an existing Virtual Machine using VMware** and testing the cloned VM by loading the previous version.

Algorithm

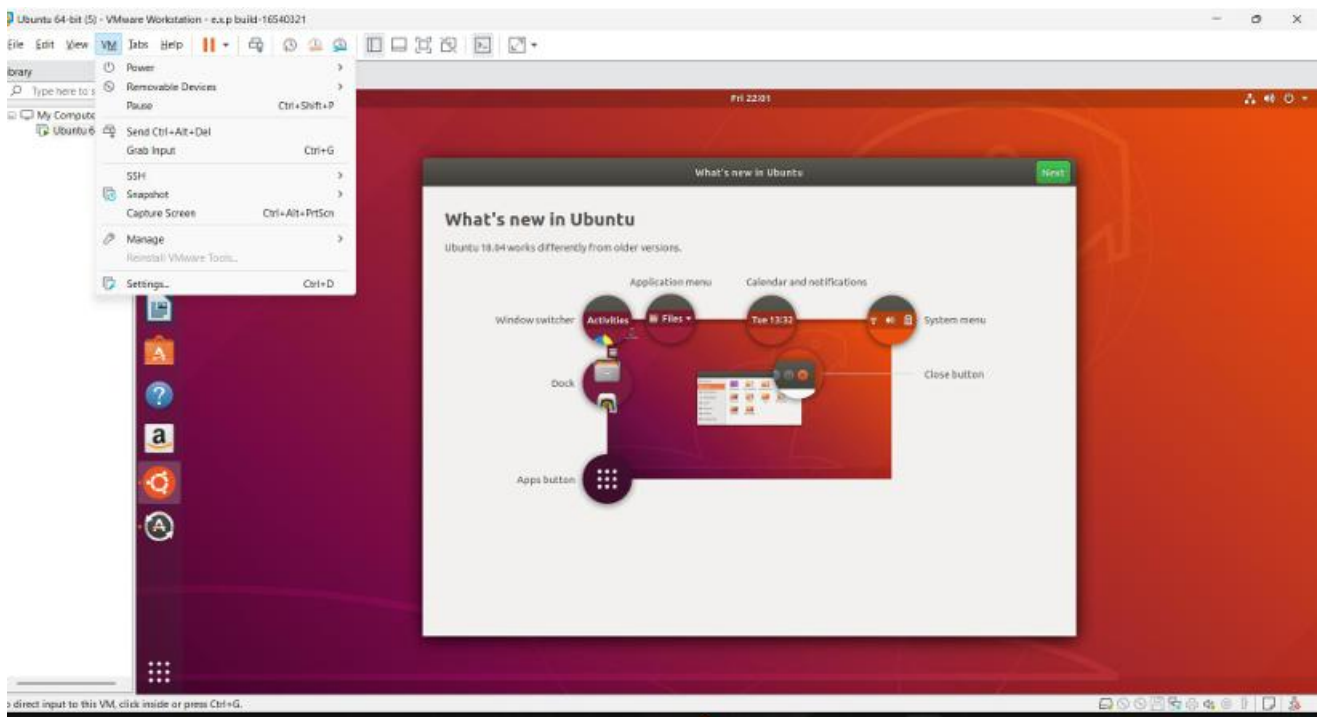
1. Open VMware Workstation.
2. Select the previously created VM.
3. Create a clone of the VM.
4. Select the appropriate cloning option.
5. Give a name and location for the cloned VM.
6. Complete the cloning process.
7. Start the cloned VM.
8. Verify the guest operating system and files.
9. Compare the cloned VM with the original VM.
10. Confirm successful cloning.

IMPLEMENTATION:

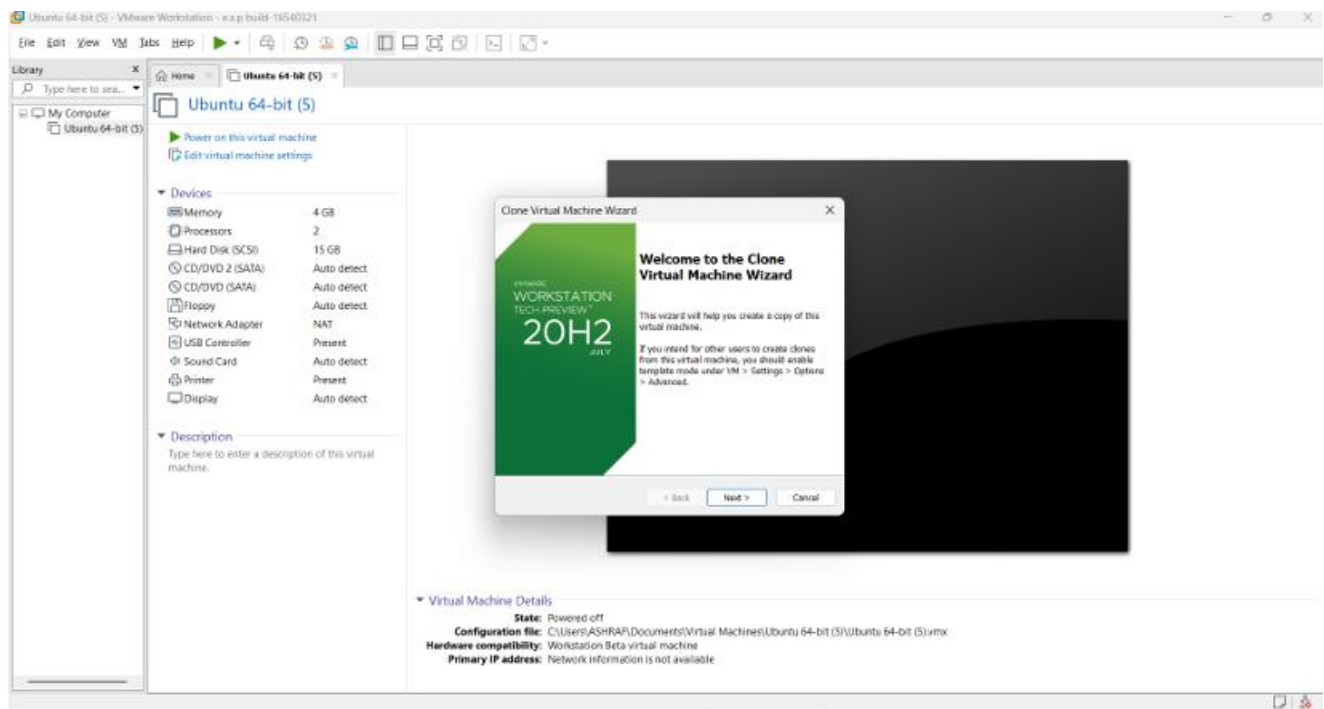
STEP 1: Open VMware Workstation, select the Ubuntu VM, and shut it down.



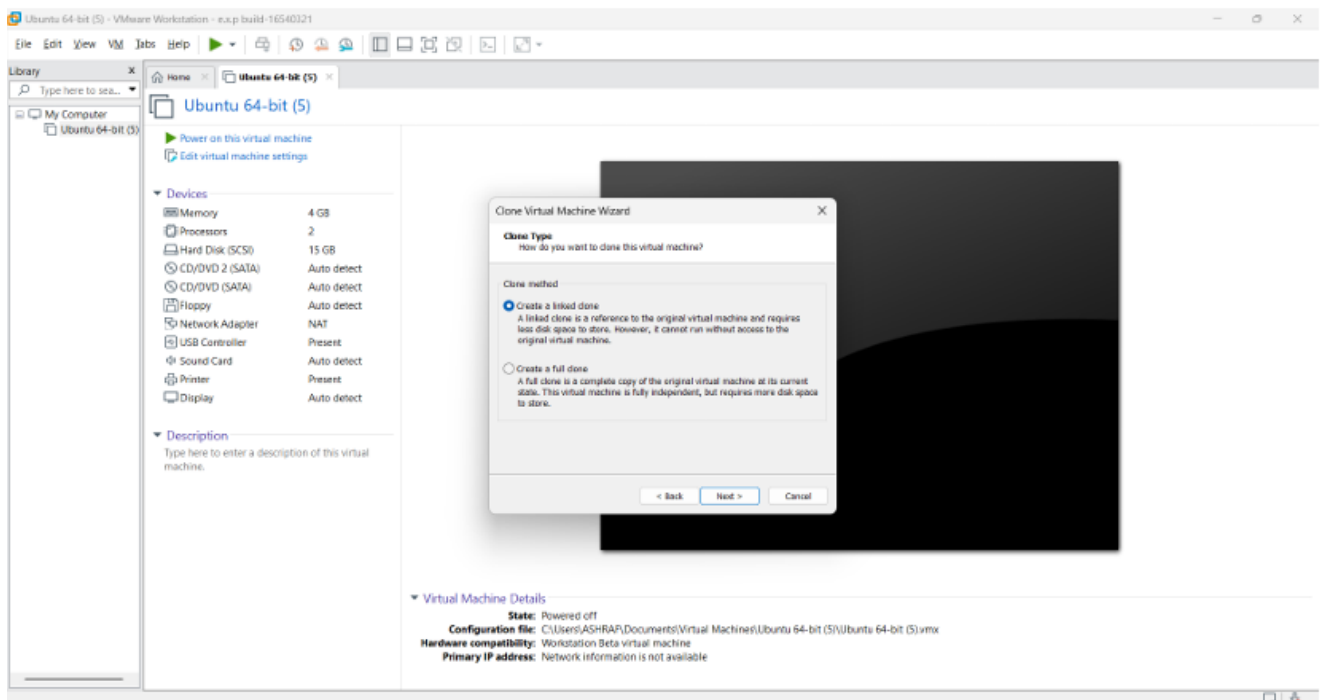
STEP 2: Select Manage → Clone to start the cloning process.



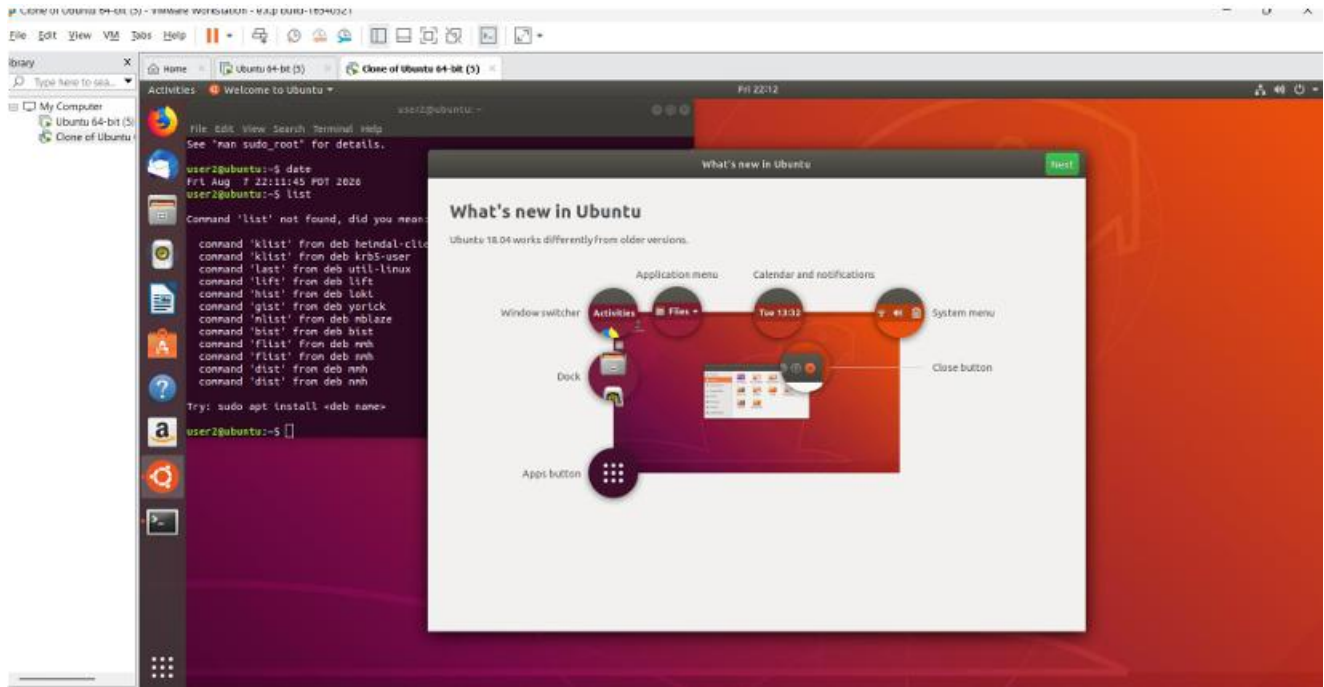
STEP 3: Select Full Clone and enter a name for the cloned VM.



STEP 4: Complete the cloning process and start the cloned VM.



STEP 5: Restore the previous snapshot, start the VM, and verify that the previous state is loaded.



Result : The VM was successfully cloned and tested, and its previous saved state was successfully restored